

Day 6 — The Pañcāṅga: Five Limbs

Module: Moon Phases & Tithi · **Band:** Middle · **Time:** 45 min

Learning objective

Students will name the five limbs (*aṅga*) of the *pañcāṅga* — tithi, vāra, nakṣatra, yoga, karaṇa — and explain in one sentence each what each limb tracks.

Materials

- The *pañcāṅga* sample page used on Day 4
- Board with a 5-row table ready
- Notebooks

Vocabulary introduced today

- *pañcāṅga* (IAST: pañca-aṅga; lit. “five limbs”) — the traditional Indian almanac listing five astronomical quantities for each day.
- *aṅga* (IAST: aṅga; lit. “limb, part”) — one of the five components.
- *vāra* (IAST: vāra; lit. “day of the week, occasion”) — the weekday.
- *yoga* (IAST: yoga; lit. “joining, combination”) — here, one of 27 astronomical divisions based on the sum of Sun and Moon longitudes. (Distinct from yoga as in Patañjali’s discipline — same word, different application.)
- *karaṇa* (IAST: karaṇa; lit. “doing, half-portion”) — a half-tithi. There are 60 karaṇas in a synodic month.

Lesson flow

Warm-up (5 min)

- Daily prompt: “*Did you see the Moon last night? What tithi?*” — 2 students share.
- “*So far, we’ve spent five days on ONE limb of the pañcāṅga — the tithi. The pañcāṅga has FOUR more limbs. Today: a tour.*”

Core (25 min)

1. **The five limbs.** Write on the board:

#	Limb	What it tracks	Number of options
1	<i>tithi</i>	Moon’s elongation from Sun	30 per month (covered Days 3–4)
2	<i>vāra</i>	day of the week	7
3	<i>nakṣatra</i>	Moon’s position against the star background	27
4	<i>yoga</i>	sum of Sun-and-Moon ecliptic longitudes	27

#	Limb	What it tracks	Number of options
5	<i>karaṇa</i>	half-tithi	11 names cycling (because of how the 60 karaṇas of a month are named)

2. **Limb-by-limb walkthrough.** Spend 3 min on each, with concrete reference to the Day 4 *pañcāṅga* page:

Vāra — the weekday (7). Named after the seven classical bodies: Sun (*Ravi*), Moon (*Soma* / *Indu*), Mars (*Maṅgala*), Mercury (*Budha*), Jupiter (*Guru* / *Bṛhaspati*), Venus (*Śukra*), Saturn (*Śani*). English Sunday/Monday/... has the same Sun/Moon/Saturn pattern — same naming logic from the same classical-period astronomy.

Nakṣatra — lunar mansion (27). The Moon takes about 27.3 days to circle the sky and return to the same star background (the *sidereal* month). Indian astronomers divided the Moon’s path into 27 segments, each ~13°20’ wide, named after the star pattern in that segment (Aśvinī, Bharaṇī, Kṛttikā, Rohiṇī, ... Revatī). The *pañcāṅga* lists which nakṣatra the Moon is in each day.

Yoga — joining (27). Add the Sun’s ecliptic longitude to the Moon’s. Divide by 13°20’. The result is the yoga number (1–27). Each has a name (Viṣkambha, Prīti, Āyusmān, ...). It’s a derived quantity, useful for timing in traditional astrology.

Karaṇa — half-tithi (11 cycling). Each tithi is split into two halves. The first 7 karaṇas of the month are *cara* (“moving”) karaṇas — Bava, Bālava, Kaulava, Taitila, Garaja, Vaṇija, Viṣṭi. They cycle 8 times through the month. The last 4 are *sthira* (“fixed”) — Śakuni, Catuṣpada, Nāga, Kīṁstughna — and appear once each near *amāvasyā*. So $8 \times 7 + 4 = 60$ half-tithis per month.

3. **Why five limbs?** Each limb encodes a different astronomical fact:

- *tithi*: position of the Moon **relative to the Sun**.
- *nakṣatra*: position of the Moon **against the stars**.
- *yoga*: combined Sun + Moon position.
- *vāra*: solar day count.
- *karaṇa*: a finer division for sub-day timing.

Together they pinpoint the astronomical situation on any given day. The *pañcāṅga* is, in effect, a compact astronomical state vector.

4. **Modern parallel.** A modern astronomy almanac (e.g. an ephemeris like NASA’s Horizons system) lists planet positions, lunar phase, day of week, eclipse data, etc. The *pañcāṅga* is exactly this kind of document — older format, same purpose. The major difference: the *pañcāṅga* also carries cultural / ritual tags (auspicious times, fast days), which a modern ephemeris does not.

Activity (10 min) — Read the *pañcāṅga*

In pairs, students take yesterday’s *pañcāṅga* page and find:

1. Today’s date row.

2. The tithi (already familiar).
3. The vāra (compare with the English weekday — does it match?).
4. The nakṣatra. Try to pronounce it.
5. The yoga and karaṇa names (these are new — just read them, no need to compute).

Each pair writes one sentence: “*Today, [date], the Moon is in [nakṣatra] nakṣatra, it is [tithi] of the [pakṣa] pakṣa, and the vāra is [day].*”

Wrap-up (5 min)

- Read aloud 2–3 students’ sentences. The class reciprocates with the meanings (“Mars day,” “Saturn day,” etc.).
- **Preview Day 7:** “*Tomorrow, the big question — why do eclipses happen on amāvasyā and pūrṇimā, but not every month?*”

Student-facing content

A pañcāṅga has five limbs. Each tracks a different astronomical fact.

Limb	What it tracks
<i>tithi</i>	Moon’s elongation from the Sun (1 of 30)
<i>vāra</i>	day of the week (1 of 7)
<i>nakṣatra</i>	Moon’s position against the stars (1 of 27)
<i>yoga</i>	sum of Sun + Moon longitudes (1 of 27)
<i>karaṇa</i>	half-tithi (1 of 11 cycling names)

Knowing all five for a given moment is like having a snapshot of where the Sun and Moon are. A modern astronomy almanac does the same job in a different format.

Homework

- Continue the Moon diary.
- For Day 7: read NASA’s published lunar phase page for the current year. Note any solar or lunar eclipses listed for this year — when, where visible. Bring 1 example.

Differentiation

- **One tier up:** Compute the yoga number for today. You need the Sun’s ecliptic longitude and the Moon’s ecliptic longitude (from any astronomy app). Add them, divide by $13^{\circ}20'$, take the integer part + 1 — that’s the yoga number. Compare with what the *pañcāṅga* says.
- **One tier down:** Memorise three limbs only — *tithi*, *vāra*, *nakṣatra*. The yoga and karaṇa terms can stay as “things I’ve heard of” for now.

Teacher notes

- The *yoga* limb is the most confusing — partly because “yoga” also means the spiritual discipline. Note this explicitly. They are different applications of the same Sanskrit root *yuj* (“to join”).
- The *karaṇa* naming is genuinely complicated. Don’t try to teach all 11 names today — just enough that students know there are half-tithis and they have names.
- This is a content-heavy day. If the class is fading, move the activity to a paired worksheet for tomorrow and use today’s time for slow walkthrough.

Citation(s) used in this lesson

- General reference: the class *pañcāṅga* source. The five-limb framework dates to early Indian astronomical literature (e.g. *Pañcasiddhāntikā* of Varāhamihira, 6th c. CE), but the modern *pañcāṅga* form is later. Paraphrased; no specific verse cited.